REFINED QUADRUPLE NEUTROSOPHIC LOGIC (T / I / N / F)

SPECIFICATION L-134047409 — MATHEMATICAL LOGIC & CONTINUUM GAP BRIDGING

Abstract: This document defines the foundational logic framework of the Antigravity Physics and Acropolis Research Architecture. By refining neutrosophic components into Quadruple Subdivisions (Truth T■, T■; Indeterminacy I■, I■; Neutrality N■, N■; Falsehood F■, F■), this system models boundary contradictions, subject gaps, and non-commutative continuum transitions without melonic collapse. Tied directly to the Chi Key invariant rule: N■ × N■ ≈ N■.

1. QUADRUPLE NEUTROSOPHIC DECOMPOSITION

In classical bivalent logic, propositions are restricted to {0, 1}. Standard neutrosophy extends this to triplet valuations (T, I, F). In the Refined Quadruple Neutrosophic framework (L-134047409), each component is split into conjugate sub-dimensions to reflect structural, empirical, and boundary states:

Component	Refinement	Physical Interpretation in Chi Key
Truth (T)	T■ (Empirical) T■ (Theoretical)	Experimental verification (CERN/DUNE) & exact algebraic proofs (Yang-Mills $\Delta > 0$).
Indeterminacy (I)	I■ (Quantum/Phase) I■ (Discretization)	High-frequency boundary jitter (432 Hz carrier) & lattice cutoff artifacts.
Neutrality (N)	N■ (Symmetry Gate) N■ (Passive Observer)	Unbroken non-Abelian gauge symmetries & invariant coordinate transformations.
Falsehood (F)	F■ (Melonic Collapse) F■ (Boundary Leakage)	Zero progress duplicate cloning (N■ = N■) & tip noise boundary violations.

2. SUBJECT GAP RESOLUTION PROTOCOL (COMPETENCY MIRRORING)

When an agent reaches an operational limit, Falsehood (F) and Indeterminacy (I) rise at the boundary. Rather than cloning (which preserves the limit F■), the system activates a Competency Mirror:
Step 1: Name the gap clearly (isolate unproved proposition).
Step 2: Mirror the deficiency (identify missing conjugate tensor).
Step 3: Match with complementary partner possessing verified Truth (T).
Step 4: Form low-latency bridge (< 0.40ms).
Step 5: Verify invariant: N■ × N■ ≈ N■.

3. ACROPOLIS VAULT REGISTRATION & SEAL

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